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Scenario 1: Hospital

id Number
name String
address String
patients List[Patient]

Patient

pid → Number
pName → String
Phone → String
DOB →String
Weight → Number
Height →Number
Age → Number
Gender → String
RelativePhone → Number
Location → String
procedures List[Procedure]

pid → 1
pName → Said
Phone → 99999999
DOB → 25/1/2001
Weight → 55
Height →170
Age → 25
Gender → Male
RelativePhone → 91919191
Location → Seeb
procedures [Procedure 1]

Procedure

proID → Number

pid → Number

drName → String

procedureDiscription String

Medicines → List [Medicine]

Potion →String

mTime → String

timeOfEnter → String

timeOfDischarge → String

dateofPro → String

cost → Number

proID → 2

pid → 1

drName → Ali

procedureDiscription → “ Nothing just want a sick Leave”

Medicines → List [Medicine]

Potion → “Twice time in day”

mTime → “After eaT”

timeOfEnter → ”10:00 AM”

timeOfDischarge → ”12:00 PM”

dateofPro → “25/1/2026”

cost → 25

Medicine

mId → Number

mName → String

Prod → String

Exp → String

SideEffect → String

Potion → String

Medicine

mId → 1

mName → P

Prod → "10/2/2025"

Exp → "10/2/2027"

SideEffect → "Sleepy"

Potion → 10 G

Room

roomNo → Number

drName → String

department → String

floorNo → Number

bedsNo → Number

roomNo → 22

drName → Ali

department → "eyesDept"

floorNo → 2

bedsNo → 6

Scenario 2: Library

members

mid → Number

mName → String

Phone → String

civilId → Number

DOB → String

Gender → String

Location → String

Books → LIST[book]

Book

Bid String

Bname String

Btype String

Quantity Number

noOfPages Number

Author String

dateOfPublish String

noOfAllowedDays String

loans → List [Loan]

Loan

Bid → String

dateOfBorrow → String

dateOfReturn → String

mid → 12

mName → Mohammed

Phone → 91111111

civilId → 141414

DOB → "12/3/2000"

Gender → "Male"

Location → "Ibri"

Books → LIST[book1,book2]

book1

Bid 1

Bname The 100

Btype "history"
Quanity 5
noOfPages 209
Author "Mohammed Professor"
dateOfPublish "20/2/2001"
noOfAllowedDays "1 month"
loans → List [Loan1]

Loan 1

Bid → 5
dateOfBorrow → "25/2/2026"
dateOfReturn → "25/3/2026"

book2
Bid 4
Bname "HowToLove"
Btype "Romance"
Quanity Number
noOfPages 321
Author "Abdulaziz"
dateOfPublish "20/2/2005"
noOfAllowedDays "1 Week"
loans → List [Loan2]

Loan 2

Bid → 5
dateOfBorrow → "30/2/2026"
dateOfReturn → "29/3/2026"

Scenario 3: Restaurant Order

Customer

Cname → String
TableNo → Number

order

orderID → Number

orderTime → String

Customization → String

items → List [item]

totalPrice → Number

Item

itemName → String

itemtype → String

dishes → List[Dish]

ItemQuantity → Number

ExpDate: → String

itemPrice → Number

tax → Number

Dish

DishType → String

DishSize → String

Customer

Cname → Said

TableNo → 101

order

orderID 12

orderTime 12:00 pm

Customization → Extra katchap

items List [item1, item2]

tax 0.05

totalPrice 1.750

Item 1

itemName Sandwich

itemtype Sandwich

dishs List[dish1]

ItemQuantity 2

ExpDate: 26/7/2026

itemPrice 0.400

Dish

DishType → Blastic

DishSize → Small

Item 2

itemName Chicken Burger

itemtype Burgers

dishs List[dish1]

ItemQuantity 1

ExpDate: 26/8/2026

itemPrice 0.800

Dish

DishType → Blastic

DishSize → Medium

flight

CompanyName → String

flightNo → String

From Country → String

To Country → String

DepresureTime → String

ArrivalTime → String

GateNo → String

OpenGate → String

booking

bookingId → String

bookingDate → String

bookingStatus → String

numberofpassengers → Number

TotalPrice → Number

passenger

passengerID→ String

Pname → String

Gender → String

BirthofDate → String

Phone → String

seat

seatNumber → String

seatClass → String

seatPossition → String

status → String

price → number

flight

CompanyName → SalamAir

flightNo → 202

FromCity → Muscat

To Country → Barcelona

DepresureTime → "2:00 Am"

ArrivalTime → "3:00 Pm"

GateNo → "G6"

OpenGate → String

bookings List [b1]

booking b1

bookingId → 12

bookingDate → "26/7/2026"

bookingStatus → "Pending"

TotalPrice → Number

passengers List [p1,p2]

passenger p1

passengerID → 1

Pname → Fatima

Gender → Female

BirthofDate → "26/8/2026"

Phone → 9111111

passengers List [s1]

passenger p2

passengerID → 1

Pname → Abdulaziz

Gender → Male
BirthofDate → "26/8/2026"
Phone → 9111111
passengers List [s2]

seat s1

seatNumber → 12
seatClass → "VIP"
seatPossition → "Windo"
status → "Available"
price → 200

seat s2

seatNumber → 13
seatClass → "VIP"
seatPossition → "middlel"
status → "Available"
price → 200

Scenario 5: Online Shop

Shop	Type
id	number
sname	String
slocation	String
Scategory	String
shoppers	List[Shopper]

DISPLAY sName → it will show shop name
Update sName → it will allow to update shop name
DisplayLocation → it will show location
UpdateLocatoin → it will allow the shop to update there location

Shopper	Type
sid	Number
Sname	String
Sphone	String
Slocation	String
SEmail	String
BankInfo	List [Bank]
purchases	List [Purchase]

Purchase	Type
PurchaseID	Number
PurchaseDate	String
TotalAmount	String
PaymentStatus	String
SEmail	String
linelItems	List [linelItem]
Delivery	delivery

linelItem	Type
ID	Number
Quantity	String
Subtotal	Number
product	String

updateQuantity()
 + calculateSubtotal()
 + displayLinelItem()

delivery	Type
ID	Number
Dname	String
Dphone	Number
Dlocation	String

updateStatus→ Change delivery stage (shipped/delivered)

estimateArrival→ Show expected delivery date

product	Type
pid	Number
pname	String
unitPrice	Number
Description	Number
StockQuantity	String

Bank	Type
Bid	Number
name	String
cardNumber	Number
CCV	Number
ExpDate	String
cardType	List [credit, dept]

Shop	Type
id	65443Ds
sname	electronis
slocation	oman
Scategory	phone
shoppers	all people

Shopper	Type
sid	6533455rSa
Sname	fatma
Sphone	98876423
Slocation	muscat
SEmail	fatma@gmail.com
BankInfo	bankmct
purchases	iphone

Purchase	Type
PurchaseID	875334tyBvckjhg
PurchaseDate	2-2-2025
TotalAmount	400OMR
PaymentStatus	succes
SEmail	fatma@gmail.com
lineltems	
Delivery	delveryemployee

lineItem	Type
ID	87654EDRFGhjk
Quantity	1
Subtotal	400OMR
product	phones

delivery	Type
ID	987654ewsdFGhjkl
Dname	suliman
Dphone	97642347
Dlocation	bahla

product	Type
pid	987654edfghjkmjnhbg
pname	phone
unitPrice	OMR
Description	iphone15pro max
StockQuantity	5

Bank	Type
Bid	09876543wsdijuygf5rf6g7h8t
name	muscat

cardNumber	09987543345677543
CCV	654
ExpDate	30-2030
cardType	credit

Scenario 6: Gym

Gym	Type
GymName	Number
phone	String
Location	String
members	List [nenber]
trainers	List [trainer]

`registerMember()` → Add new member.

`addTrainer` → Add trainer to gym.

`scheduleClass` → Create class session.

member	Type
Mid	String
Mname	String
Phone	String
DOB	String
gender	String
memberships	membership

`enrollClass` → Join a fitness class. `renewMembership` → Extend membership.

membership	Type
mbid	Number
typeOfSubscription	String
startDate	String
lastDate	String
cost	String

trainer	Type
id	Number
name	String
email	String
phone	String
DOB	List [trainer]
Experience	String
gender	String

markAttendance → Record who attended.

updateSchedule → Change class timing.

class	Type
id	Number
TrainingType	String
scheduledTimes	List [scheduledTime]

scheduledTimes	Type
day	Number
startTime	String

endTime	String
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-----Example-----

Gym	Value
GymName	WAWN
phone	999999999
Location	Muscat
members	List [member1]
trainers	List [trainer1]

member1	Value
Mid	1
Mname	Ali
Phone	99990000
DOB	20/1/2001
Gender	Male
memberships	membership

membership	Value
mbid	2

typeOfSubscription	VIP
startDate	20/1/2028
lastDate	20/1/2029
cost	300

trainer1	Value
id	2
name	Said
email	example@gmail.com
phone	99999999
DOB	20/1/2001
Experience	20 Years
gender	Male
classes	List [class1,class2]

class1	Type
id	1
Cname	"How to be strongest one in the Gym"
TrainingType	"Arm Training"
scheduledTimes	List [scheduledTime1,scheduledTime2]

scheduledTimes 1	Value
day	Monday
startTime	8:00 PM

endTime	10:00 PM
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Scenario 7: Music Playlist App

user	type
userid	Number
name	string
Email	string
phone	string
Playlists	List [playlist]

`createPlaylist` → Create playlist.

`deletePlaylist` → Remove playlist.

playlist	type
playlistid	string
titlelist	string
createddate	string
Albums	List [Album]

`addTrack` → Add song.

`removeTrack` → Remove song.

`reorderTracks` → Change song order.

Album	type
albumid	Number
albumname	String

relaseyear	String
artist	String

Track	type
tarckid	string
title	string
duration	string

play → Start track.

getDuration → Return song length.

-----Example-----

user	type
userid	98765Fd
name	nasser
Email	naser@email.com
phone	98765445
Playlists	sad

playlist	type
playlistid	sad
titlelist	sadday
createddate	2-2-2010
Albums	sadness

Album	type
albumid	23456Db
albumname	sadness
relaseyear	2010
artist	jack

Track	type
tarckid	0987642KjhKMJHGGFRt
title	sadness
duration	3:54

Scenario 8: Construction Project

company	Type
id	Number
cname	String
location	String
projects	List [Project]

project	Type
id	Number
pName	String
startdate	String
endDate	String
Cost	Number
Phases	List [phase]

addPhase(→ Add project phase.
calculateCost → Sum all task costs.

phase	Type
id	Number
pName	String
startdate	String
endDate	String
Cost	Number
tasks	List [task]

ddTask → Add task to phase.
checkProgress → Show completed tasks.

task	Type
id	Number
pName	String
startdate	String

endDate	String
Cost	Number
workers	List [worker]

assignWorker→ Link worker to task.

markCompleted→ Set task finished.

worker	Type
id	Number
name	String
email	String
salary	Number
phone	String
gender	String
nationality	String
role	String

-----**Example**-----

company	Type
id	236Sa
cname	legend
location	oman-muscat
projects	coding

project	Type
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id	2345643Nb
pName	site
startdate	21-5-2026
endDate	22-7-2026
Cost	10000 OMR
Phases	second phase

phase	Type
id	28765Sd
pName	build departemnt
startdate	4-6-2026
endDate	30-6-2026
Cost	2000OMR
tasks	4 task]

task	Type
id	87654DS
pName	finish the type department
startdate	10-6-2026
endDate	1-7-2026
Cost	1500 OMR
workers	1

worker	Type
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id	8765Ds
name	mohammed
email	mohed@email.com
salary	2000OMR
phone	999999999
gender	male
nationality	omani
role	build site

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[Scenario 9: University Department]

Programme	type
id	number
programmeName	string
degreeType	string
durationYears	string
semesters	List[Semester]

addSemester → Add semester.

countModules → Count all modules.

Semester	Type
id	Number
semesterNumber	Number
academicYear	String
modules	list[modules]

addModule→ Add module.

module	type
id	Number
moduleCode	string
moduleName	string
credits	Number
assessments	List[Assessment]

addAssessment → Add exam/assignment.

calculateTotalMarks → Sum assessment percentages.

Assessment	type
id	number
title	string
assessmentType	string
Percentage	number
dueDate	string

updateDeadline → Change due date.

-----Example-----

Programme	type
id	GH-101
programmeName	math
degreeType	bacholer
durationYears	5
semesters	10

Semester	Type
id	ed-320
semesterNumber	2
academicYear	2023
modules	4]

module	type
id	kh-101
moduleCode	876
moduleName	integration
credits	23
assessments	3]

Assessment	type
id	78765Z
title	loop
assessmentType	MQ
Percentage	100
dueDate	21-2-2024

Scenario 10: Car Rental

company	Value
ID	Number
cname	String
Location	String

vehicles	List [vehicle]
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assignCustomer → Link customer to rental.

calculateCost → Compute rental fee.

closeRental → Finish rental.

vehicle	Value
id	Number
Vname	String
color	String
plate number	String
model	String
brand	String
year	Number
serviceshistory	maintenanceRecord
mileage	String

updateMileage → Save new mileage.

addMaintenanceRecord → Add service history.

rental	Value
ID	Number
Customer	customer
Vehicle	vehicle
Duration	String
StartDate	String

maintenanceRecord	Value
ID	Number
workshopname	String

Date	String
Vehicle	vehicle
Description	String
nextService	String
Cost	String

`scheduleNextService` → Set next service date.

`describeService` → Show maintenance details.

customer	Value
ID	number
fullname	string
phoneNumber	string
driveLicense	Bool

-----Example-----

company	Value
ID	21
cname	Happy Rental
Location	Al-Seeb
vehicles	List [vehicle1]

vehicle 1	Value
id	21

Vname	Kia
color	Blue
plate number	3218
model	Optima
brand	Kia
year	2027
serviceshistory	maintenanceRecord
mileage	74.138

rental	Value
ID	1
Customer	customer1
Vehicle	vehicle1
Durition	20 Days
StartDate	10/02/2027

maintenanceRecord	Value
ID	22
workshopname	Happy Service
Date	20/08/2027
Vehicle	vehicle 1
Description	Every thing work fine only major service
nextService	20/12/2027
Cost	12.000

customer1	Value
ID	1
fullname	Said Al Sharji
phoneNumber	99999999
driveLicense	yes